

PHYSICS EXAM-17 : URGENT MARKING REPORT

Unit 7 Density and Temperature, Parts 7A and 7B (27 marks) + Revision: Unit 6 Work and Energy, Unit 3 Dynamics-I (11 marks)
Seemab · Class 9 · Attempt Date: 2026-09-25 · Paper: 38 marks, 1h30m, no optional questions, Version A

Final Score: 36.5 / 38 · 96.1% · Grade A

Marking notes. All three sections are present and the question sequence is continuous across the 7-page CamScanner scan (p1 Q.2, p2 Q.3 and Q.4(i), p3 Q.4(ii), p4 Q.5(A), p5 Q.5(B), p6 Q.6(i) to (iii) and the start of (iv), p7 the rest of Q.6(iv) and the MCQ list). Nothing is missing and there is no gap suggesting an unscanned page. She wrote her MCQ answers as a plain numbered list on the last page, her usual habit. A blind second marker worked the same script without seeing the first pass and returned the identical total of 36.5, the identical two deductions and the identical section splits. Every correct answer was re-derived from her own textbook rather than taken from the key. **This paper carried untaught material** (Unit 7 Parts 7A and 7B had not been covered in class), so the score is split into new material and revision below.

10 / 10
SECTION A (MCQS)

11.5 / 12
SECTION B (SHORT)

15 / 16
SECTION C (LONG)

25.5 / 27
UNIT 7, NEW MATERIAL (94.4%)

11 / 11
REVISION (100%)

SECTION A : Multiple Choice Questions (10 / 10)

#	Question (summary)	Correct	Seemab's	Result
1	2700 kg/m ³ written in g/cm ³	C : 2.7 g/cm ³	C	1
2	Density of a cube, side 3 cm, mass 81 g	A : 3 g/cm ³	A	1
3	According to the book, density is	B : a scalar quantity	B	1
4	Volume of a gas in its container	D : equal to the container	D	1
5	Liquids and gases together are called	C : fluids	C	1
6	Why plasma conducts current well	B : free electrons and moving ions	B	1
7	Heated gas atoms losing electrons become	A : positive ions	A	1
8	[Revision, U6] Mass halved, speed doubled	D : 2 E _k	D (self-corrected)	1
9	[Revision, U6] Energy stored by dam water	C : gravitational PE	C	1
10	[Revision, U3] Mass of a stone weighing 540 N	B : 54 kg	B	1

A clean sweep, and the shift guard confirms it. Her list was tested against the key at offsets 0, +1, −1, +2 and −2. Offset 0 gives 10 of 10; every other alignment gives zero matches anywhere. There is no reading artefact here, the ten are simply right. All ten are numbered in her own hand with none missing.

MCQ 8 is the one to celebrate. A letter is scribbled through and D written clearly beside it. This is the half-mass-double-speed item that cost her a mark in the OPF First Term Physics paper, deliberately re-served here. She started down the wrong path, caught herself and corrected it unprompted. The trap option (c) 4 E_k, which ignores the halved mass, did not catch her this time.

SECTION B : Short Questions (11.5 / 12)

Q.2 : Density definition, equation, volume formulas 4 / 4

"Density: mass per unit volume of the substance is called density. Formula/Equation: Density = Mass/Volume or $\rho = m/V$. It's SI unit stands as kilogram per cubic meter (Kg/m³). Smaller unit: gram per cubic centimeter (g/cm³)"

"Volume of cuboid :- $V = l \times w \times h$. Volume of cube :- $V = a^3$. Sphere : $V = 4/3 \pi r^3$ "

"The 2 cubes have same density bcuz they are made of the same material, iron. The large cube has greater mass and volume in the same proportion, so mass per unit volume remains unchanged."

Full marks. The definition is the book's own sentence word for word, the equation is given in both word and symbol form, the SI unit is named, and all three volume formulas are right with the sphere correctly written in r and not d, which is where this item usually leaks half a mark. Part (iii) is the proportionality argument in her own words and it is exactly the idea the book states. This is Quiz 1 and Quiz 3 from her Part 7A notes, both banked.

Q.3 : Density of the liquid, 120 g empty, 50 mL, 160 g total 4 / 4

"mass of liquid = total mass – mass of cylinder. 160 – 120 = 40g"

" $\rho = 40/50 \Rightarrow 0.8 \text{ g/ml}$ · convert to Kg/m³ : 1 g/ml $\Rightarrow 1000 \text{ Kg/m}^3$. $0.8 \times 1000 = 800 \text{ Kg/m}^3$ "

"full sentence :- The density of liquid is 800 Kg/m³"

Full marks, and the cleanest question on the paper. She named the subtraction step before doing it, wrote the formula before substituting, carried the unit at every stage and closed with the full sentence the paper asked for. This is Quiz 2 from her Part 7A notes with the exact numbers she had practised, and she took all four marks.

Q.4 : The three states of matter 3.5 / 4

	Solid	Liquid	Gas
Arrangement	"Closly packed, tightly packed."	"Packed but Passing."	"flowing"
Movement	"It is in mean position but vibrate."	"Their position change."	"Random motion."
Density	"High"	"medium (intermediate)"	"Low"

Eight of the nine cells accepted, one rejected. Each cell was judged on meaning against Table 7.2 and section 7.3.1, not on matching the book's words. "Closly packed" is the book's own phrase for solids. "Their position change" is essentially the book's clause for liquids. "Packed but Passing" is unusual wording but it carries both halves of what the book says about liquid arrangement, close together yet gliding past one another, and it places the liquid correctly between the solid and the gas on the axis the heading asks about, so it is credited.

The half mark is lost on arrangement for a gas, where she wrote "flowing". That describes how a gas moves, not how its particles are arranged, and it fails to separate a gas from a liquid at all, since the book's own Point to Ponder groups the two together precisely because both flow. She had already used her movement cell for motion. The word she needed, **random**, was sitting one row below in her own table.

"The attractive forces between particles in a solid are strong, so the particles are held very close together. In a gas, the attractive forces are very weak, so the particles are far apart. Therefore, a solid has much more mass in the same volume and hence has a much higher density than a gas."

Part (ii) is full marks and genuinely well written: both sides of the comparison, the force stated first and the density conclusion drawn from it. This is Quiz 3 from her Part 7B notes.

SECTION C : Long Questions (15 / 16)

Q.5 : Finding the density of solids 7 / 8

"Initial water level = 30 mL. final water level = 55 mL. $V = 55 - 30 = 25 \text{ ml}$. Since 1 ml = 1 cm³ = 25 cm³"

" $\rho = m/V = 175/25 = 7 \text{ g/cm}^3$ "

"method :- The method is the water displacement method. It is used for an irregularly shaped solid. The object must sink completely in water. It must not dissolve / react with water."

Part (A) is perfect, 4 out of 4. The volume conversion is stated rather than assumed, the formula precedes the substitution, the unit is present, and part (iii) gives the method, the kind of object and both of the book's conditions.

"20cm = 0.20m, 10cm = 0.10m, 5cm = 0.05m. Volume :- $V = l \times w \times h = 0.20 \times 0.10 \times 0.05 = 0.001 \text{ m}^3$. Density :- mass = 19.3 Kg. $\rho = m/V = 19.3/0.001 = 19300 \text{ Kg/m}^3$ "

Parts (iv) and (v) are also perfect, 2.5 out of 2.5. Every length converted, the volume right, the division right, the unit written.

"(vi) metal – The metal is tungsten, bcuz it's density is approxiam-etdy 19300 kg/m³. full sentence : The metal is tungsten."

This is the one real loss on the paper, and it deserves explaining rather than just marking. Her physics is not wrong about the world. Tungsten really does sit at about 19250 kg/m³, which is why it is the classic material for faking gold bars. But the question said **"Using Table 7.1 of your book, name the metal"**, and Table 7.1 was read cell by cell to settle this: it lists iron 7900, gold 19300, ice 920, plythene 900, petrol 800, pure water 1000, mercury 13600, air 1.3 and carbon dioxide 2.0. **Gold is the only entry at 19300, and tungsten is not in the table at all.** She got the number exactly

right and then went outside the book for the name instead of opening the table the question pointed her to. The 1 mark for the name is lost.

The separate half mark for the closing sentence stands. The paper prices that sentence on its own, instruction 7 requires it as a written habit, and she wrote a complete sentence answering the question and even labelled it "full sentence" to show she knew the rule. Withholding it would charge her twice for one error, and this paper's own rule is that marks follow the working: she carried her own answer forward correctly. Both markers agreed on this independently.

Q.6 [Revision, Unit 6] : Efficiency 8 / 8

"Efficiency is the ratio of useful output energy to total input energy, usually expressed as percentage. Percentage Efficiency = useful output energy / input energy x 100"

"Efficiency = 1500/2500 x 100 = 60%"

" $P_{out} = \eta \times P_{in}$. Efficiency as a decimal: $\eta = 60\% = 0.60$. $P_{out} = 0.60 \times 600$. $P_{out} = 360\text{ W}$ "

"A machine cannot have 100% bcuz some of the input energy is always wasted, for example as heat or sound, due to friction and other energy losses. and the name of the device which have lowest efficiency: Incandescent Ligth $\Rightarrow$ 5%"

Eight out of eight, every part of it. The trap in (iii) is using 60 instead of 0.60, which gives 36,000 W; she wrote out the conversion to a decimal as its own labelled step before substituting, which is exactly the discipline that prevents it. Table 6.3 was checked independently and the incandescent light at 5% is right.

RE-TEST OUTCOME (deliberately re-served from earlier papers)

Re-test	Result	Evidence
MCQ 8, mass halved and speed doubled (OPF First Term Physics MCQ 9)	FIXED	Struck out a first letter, settled on D, which is correct. Self-caught with no prompting.
MCQ 9, energy stored by dam water (OPF First Term Physics MCQ 11, where she chose electrical)	FIXED	Answered C, gravitational potential energy.
MCQ 10, weight treated as mass (OPF First Term Physics 2(ii))	FIXED	$540 \div 10 = 54\text{ kg}$. Did not write 540 as the mass.
Missing concluding sentence (dropped across six earlier papers)	FIXED	Written on both questions that asked for one, Q.3(iii) and Q.5(vi), and labelled "full sentence" both times.
Missing units (recurring half-mark losses)	FIXED	Every numerical answer on the paper carries its unit. Zero unit deductions.

WHAT THIS PAPER SHOWS

The notes did their job. All six quiz questions from her Aasan Notes Parts 7A and 7B were on this paper, worth 13 of the 27 new marks. She took all 13. On material that had not been taught in class, she scored 25.5 out of 27.

Three long-standing habits are now fixed at once. Units, concluding sentences and the three First Term MCQ errors were all re-served deliberately on this paper and every one of them came back correct. The concluding sentence in particular had been missing across six papers; here she wrote it twice and labelled it.

The one thing to work on is small and specific. When a question names a table, open that table. She lost a mark not because she did not know the physics but because she answered from memory when the book was sitting in front of her. The same instinct is worth watching in the states-of-matter table, where she gave a movement word in an arrangement box.